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Teaching statistical learning at university

A 14-week, 3-credit module for second- or third-year students in statistics, data science or a quantitative social science. Prerequisites: introductory probability and one programming language (Python or R). Darsi's four lectures fall in weeks 2, 7, 9 and 11; the staged project runs through weeks 11–14.

Learning outcomes

1. Explain a sampling distribution and compute standard errors analytically and by bootstrap.
2. Interpret confidence intervals correctly and identify common misreadings.
3. Diagnose bias and variance and estimate test error with leak-free cross-validation.
4. Justify a causal adjustment set from a DAG and estimate an effect by difference-in-differences.
5. Produce a reproducible analysis of real data that states a claim, its uncertainty and its limits.

14-week syllabus

Week	Topic	In class and out
1	Data, populations and samples	Lecture; sampling-bias case studies
2	Sampling distributions by simulation	Darsi lecture 1; sampling lab
3	Standard error and the CLT	Derivation; problem set 1 out
4	Confidence intervals and the bootstrap	Bootstrap lab in Python or R
5	Hypothesis tests and their misuse	Seminar on p-values; quiz 1
6	Linear regression as prediction	Lecture; problem set 2 out
7	Bias, variance and model complexity	Darsi lecture 2; overfitting lab
8	Cross-validation and leakage	Find-the-leak seminar
9	Confounding and causal diagrams	Darsi lecture 3; DAG workshop
10	Difference-in-differences	Case study; quiz 2
11	Project proposal	Darsi lecture 4; proposal due
12	Reproducible workflows	Version-control and notebook clinic
13	Project draft and peer review	Rubric-based peer feedback
14	Final presentations	Presentations; final report due

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Lecture slide outline

Lecture 1: Sampling distributions and the bootstrap

- Hook: what does 95% mean? (poll)
- Simulate 1,000 samples, n = 10 and 40
- Derive $\mathrm{Var}(\bar{x}) = \sigma^2/n$; the CLT
- Efron's bootstrap; percentile intervals
- When the bootstrap fails

Lecture 2: Bias, variance and cross-validation

- Hook: 99% accuracy, on what?
- Decomposition: $\text{bias}^2 + \text{variance} + \sigma^2$
- The U-curve, live with polynomials
- k-fold CV and a final test set
- Leakage and pipelines

Lecture 3: Causal inference

- Hook: close the tutoring centre?
- Potential outcomes; randomisation
- DAGs: confounders, mediators, colliders
- The back-door criterion
- Difference-in-differences and parallel trends

Lecture 4: Designing the module

- Constructive alignment
- GAISE recommendations
- Staged projects on real data
- Reproducibility as a requirement
- Rubrics and feedback on drafts

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Problem set

1. A population has σ = 15. Find the SE of the mean for n = 25 and n = 225, and explain the ratio.

2. Write pseudocode for a percentile bootstrap 95% interval for a median, with B = 2,000.

3. A student reports: “There is a 95% probability that μ lies in [4.1, 5.3].” Write feedback that corrects this in two sentences.

4. Fold errors from 5-fold CV are 0.21, 0.19, 0.25, 0.22, 0.18. Give the CV estimate and one reason it may still be optimistic.

5. In the DAG Z → X → Y, Z → Y, X → M → Y, X → C ← Y, which variables should be adjusted for to estimate the total effect of X on Y?

6. Treated: 62 → 71; comparison: 60 → 64. Compute the DiD effect and describe one check of its key assumption.

Solutions

> **1.** 15/5 = 3 and 15/15 = 1. Nine times the data divides the SE by √9 = 3.

> **2.** Repeat 2,000 times: resample n values with replacement and store the median. Sort the 2,000 medians; the interval is from the 50th to the 1,950th value.

> **3.** μ is fixed, so this interval either contains it or not. The 95% means that the method captures μ in about 95% of repeated samples.

> **4.** Mean = 1.05/5 = 0.21. It is optimistic if any preprocessing or model selection used all the data, or if related rows were split across folds.

> **5.** Adjust for Z only: it is a confounder. M is a mediator and C a collider; adjusting for either biases the total effect.

> **6.** (71 - 62) - (64 - 60) = 9 - 4 = 5. Plot several pre-treatment periods and check the two groups moved in parallel.

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Lab handout: Bootstrap and leakage

Goal

Estimate uncertainty without a formula, and see how leakage inflates a cross-validated score.

Steps

1. Load a public dataset with at least 500 rows (for example, a national household survey extract).
2. Bootstrap the median and a correlation with B = 2,000; report SEs and percentile intervals.
3. Compare the bootstrap SE of the mean with $\sigma/\sqrt{n}$.
4. Fit a model with feature selection done on all rows, then inside a Pipeline; compare the 5-fold CV scores.
5. Write a one-paragraph interpretation of one interval and one explanation of the score gap.

Hand in

A notebook that reruns from raw data with one command, plus the two paragraphs.

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Assessment rubric

Criterion	Excellent	Good	Developing	Not yet
Statistical reasoning	Correct methods, assumptions stated and checked	Correct methods, assumptions stated	Methods partly correct	Methods missing or wrong
Interpretation	Claims match evidence, uncertainty and limits explained	Claims match evidence, uncertainty given	Some overclaiming	Claims not supported
Reproducibility	Reruns with one command, clean history	Reruns with minor fixes	Partly reruns	Does not rerun
Communication	Clear, well-chosen figures, concise	Clear with minor issues	Hard to follow in places	Unclear

Reading list

Wasserman, L. (2004). All of Statistics. Springer.
Core text for sampling distributions and the bootstrap.

Efron, B. (1979). Bootstrap methods: another look at the jackknife. Annals of Statistics, 7(1), 1–26.
The original bootstrap paper.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.
Bias–variance and cross-validation, chapter 7.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning (2nd ed.). Springer.
Accessible student text with labs.

Pearl, J. (2009). Causality (2nd ed.). Cambridge University Press.
DAGs and the back-door criterion.

Angrist, J. D., & Pischke, J.-S. (2009). Mostly Harmless Econometrics. Princeton University Press.
Difference-in-differences.

GAISE College Report ASA Revision Committee (2016). Guidelines for Assessment and Instruction in Statistics Education College Report. American Statistical Association.
Recommendations for teaching statistics at university.

Biggs, J., & Tang, C. (2011). Teaching for Quality Learning at University (4th ed.). Open University Press.
Constructive alignment.

Freeman, S., et al. (2014). Active learning increases student performance in science, engineering, and mathematics. PNAS, 111(23), 8410–8415.
Evidence for active learning in STEM.

Hattie, J., & Timperley, H. (2007). The power of feedback. Review of Educational Research, 77(1), 81–112.
Feedback that says where to go next.